

Homothetic and Nonhomothetic CES

Study notes

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1 Homotheticity

1.1 Homothetic preferences

Let the consumption vector be $\mathbf{c} = (C_1, \dots, C_I) \in \mathbb{R}_{++}^I$, and let $\succeq$ denote the consumer's preference relation. Preferences are *homothetic* if, for any consumption vectors $\mathbf{c}, \mathbf{d}$ and any $a > 0$,

$$\mathbf{c} \succeq \mathbf{d} \iff a\mathbf{c} \succeq a\mathbf{d}. \quad (1.1)$$

Scaling both consumption bundles by the same positive factor does not change the consumer's preference ranking.

1.2 Homothetic CES

Consider I goods and define the consumption aggregate by

$$C = F(C_1, \dots, C_I) = \left[\sum_{i=1}^I C_i^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad C_i > 0, \quad \sigma \in (0, 1) \cup (1, \infty). \quad (1.2)$$

Here C_i is the consumption quantity of good i , C is the consumption index aggregated from the individual goods, and σ is the elasticity of substitution. To be consistent with Mestieri's derivation notes, we use the form without preference weights.

By definition,

$$C^{\frac{\sigma-1}{\sigma}} = \sum_i C_i^{\frac{\sigma-1}{\sigma}} \implies \sum_i \left(\frac{C_i}{C} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (1.3)$$

This is the special case of the nonhomothetic specification below in which $\varepsilon_i = 1$ for all i .

1.2.1 Constant elasticity of substitution

Differentiating the CES aggregate gives

$$\frac{\partial F}{\partial C_i} = \left[\sum_j C_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} C_i^{-\frac{1}{\sigma}} = C^{\frac{1}{\sigma}} C_i^{-\frac{1}{\sigma}}. \quad (1.4)$$

Define the marginal rate of substitution of good i for good j as

$$\text{MRS}_{ij} = \frac{\partial F / \partial C_i}{\partial F / \partial C_j} = \left(\frac{C_i}{C_j} \right)^{-\frac{1}{\sigma}}. \quad (1.5)$$

At an interior optimum, $\text{MRS}_{ij} = P_i / P_j$, implying

$$\frac{P_i C_i}{P_j C_j} = \left(\frac{P_i}{P_j} \right)^{1-\sigma}. \quad (1.6)$$

In this homothetic CES demand system, goods are gross substitutes when $\sigma > 1$ and gross complements when $0 < \sigma < 1$. The elasticity of substitution is

$$\boxed{-\frac{d \ln(C_i/C_j)}{d \ln \text{MRS}_{ij}} = \sigma.} \quad (1.7)$$

1.3 Utility maximization

Given prices and total expenditure E , the consumer solves

$$\max_{\{C_i > 0\}} F(C_1, \dots, C_I) \quad \text{subject to} \quad \sum_i P_i C_i = E. \quad (1.8)$$

The Lagrangian is

$$\mathcal{L} = F(C_1, \dots, C_I) + \mu \left(E - \sum_i P_i C_i \right). \quad (1.9)$$

The first-order conditions imply

$$\frac{\partial F}{\partial C_i} = \mu P_i \implies \left(\frac{C}{C_i} \right)^{1/\sigma} = \mu P_i \implies C_i = C(\mu P_i)^{-\sigma}. \quad (1.10)$$

Substituting into the CES constraint,

$$1 = \sum_i \left(\frac{C_i}{C} \right)^{\frac{\sigma-1}{\sigma}} = \sum_i (\mu P_i)^{1-\sigma} = \mu^{1-\sigma} \sum_i P_i^{1-\sigma}. \quad (1.11)$$

Define the price index

$$P = \left(\sum_i P_i^{1-\sigma} \right)^{\frac{1}{1-\sigma}}. \quad (1.12)$$

Then $(\mu P)^{1-\sigma} = 1$, and hence $\mu = 1/P$. Therefore,

$$C_i = C \left(\frac{P_i}{P} \right)^{-\sigma}. \quad (1.13)$$

The budget constraint becomes

$$E = \sum_i P_i \left[C \left(\frac{P_i}{P} \right)^{-\sigma} \right] = C P^\sigma \sum_i P_i^{1-\sigma} = C P^\sigma P^{1-\sigma} = C P. \quad (1.14)$$

Thus $C = E/P$, and Marshallian demand is

$$\boxed{C_i(\mathbf{P}, E) = \frac{E}{P} \left(\frac{P_i}{P} \right)^{-\sigma} = E P_i^{-\sigma} P^{\sigma-1}.} \quad (1.15)$$

1.4 Expenditure shares and elasticities

The expenditure share of good i is

$$s_i \equiv \frac{P_i C_i}{E} = \left(\frac{P_i}{P} \right)^{1-\sigma} = \frac{P_i^{1-\sigma}}{\sum_j P_j^{1-\sigma}}, \quad \sum_i s_i = 1. \quad (1.16)$$

At fixed prices,

$$\left. \frac{\partial s_i}{\partial E} \right|_{\mathbf{P}} = 0. \quad (1.17)$$

Define the expenditure elasticity of good i by

$$\eta_i \equiv \left. \frac{\partial \ln(P_i C_i)}{\partial \ln E} \right|_{\mathbf{P}}. \quad (1.18)$$

Taking logarithms of demand gives

$$\ln C_i = \ln E - \sigma \ln P_i + (\sigma - 1) \ln P, \quad (1.19)$$

$$\ln(P_i C_i) = \ln E + (1 - \sigma) \ln \left(\frac{P_i}{P} \right). \quad (1.20)$$

Consequently,

$$\boxed{\eta_i = 1.} \quad (1.21)$$

Interpretation. Additional total expenditure is allocated in proportion to the original expenditure shares.

2 Nonhomotheticity

2.1 Definition and substitution elasticity

Nonhomothetic CES defines the consumption or utility index $C = F(C_1, \dots, C_I)$ implicitly through

$$\boxed{\sum_{i=1}^I \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = 1.} \quad (2.1)$$

Here C_i is the quantity of good i , C is the consumption or utility index, $\varepsilon_i > 0$ parametrizes the income elasticity of sector i , and $\sigma \in (0, 1) \cup (1, \infty)$ is the constant elasticity of substitution. If $\varepsilon_i = 1$ for all i , the specification becomes standard homothetic CES.

Write $\alpha = (\sigma - 1)/\sigma$. The defining equation is

$$\sum_k C_k^\alpha C^{-\alpha \varepsilon_k} = 1. \quad (2.2)$$

Along the same indifference curve, adjust only C_i and C_j , holding the quantities of all other goods constant. Since C is fixed, $dC = 0$, so

$$\frac{\partial C}{\partial C_i} dC_i + \frac{\partial C}{\partial C_j} dC_j = 0. \quad (2.3)$$

Differentiating the defining equation at fixed C also gives

$$\alpha C_i^{\alpha-1} C^{-\alpha \varepsilon_i} dC_i + \alpha C_j^{\alpha-1} C^{-\alpha \varepsilon_j} dC_j = 0. \quad (2.4)$$

It follows that

$$\text{MRS}_{ij} = \frac{\partial C / \partial C_i}{\partial C / \partial C_j} = - \frac{dC_j}{dC_i} = \frac{C_i^{\alpha-1} C^{-\alpha \varepsilon_i}}{C_j^{\alpha-1} C^{-\alpha \varepsilon_j}} \quad (2.5)$$

$$= \left(\frac{C_i}{C_j} \right)^{-1/\sigma} C^{\frac{\sigma-1}{\sigma}(\varepsilon_j - \varepsilon_i)}. \quad (2.6)$$

Taking logarithms,

$$\ln \text{MRS}_{ij} = -\frac{1}{\sigma} \ln \left(\frac{C_i}{C_j} \right) + \frac{\sigma-1}{\sigma} (\varepsilon_j - \varepsilon_i) \ln C. \quad (2.7)$$

Thus, along the indifference curve,

$$\boxed{- \frac{d \ln(C_i/C_j)}{d \ln \text{MRS}_{ij}} \Big|_C = \sigma.} \quad (2.8)$$

2.1.1 Domain and comparison with Stone–Geary preferences

Under the stated parameter restrictions, nonhomothetic CES is well-defined for all strictly positive consumption quantities. In contrast, Stone–Geary preferences,

$$U = \prod_i (C_i - \gamma_i)^{w_i}, \quad \gamma_i > 0, \quad w_i > 0, \quad \sum_i w_i = 1, \quad (2.9)$$

require $C_i > \gamma_i$ for every i in their interior domain. A more general implicit form, discussed by Comin, Lashkari, and Mestieri [1], replaces C^{ε_i} by a good-specific function:

$$\sum_i \left(\frac{C_i}{g_i(C)} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (2.10)$$

2.2 Cost minimization

Given prices $P_i > 0$ and a consumption index $C > 0$, choose the quantities of goods to solve

$$\min_{\{C_i > 0\}} \sum_{i=1}^I P_i C_i \quad \text{subject to} \quad \sum_{i=1}^I \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (2.11)$$

The Lagrangian is

$$\mathcal{L} = \sum_i P_i C_i + \lambda \left[1 - \sum_i \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} \right]. \quad (2.12)$$

The first-order conditions give

$$\frac{\partial \mathcal{L}}{\partial C_i} = P_i - \lambda \frac{\sigma-1}{\sigma} \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{-1/\sigma} \frac{1}{C^{\varepsilon_i}} = 0, \quad (2.13)$$

$$P_i C_i = \lambda \frac{\sigma-1}{\sigma} \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}}. \quad (2.14)$$

Summing over goods and using the constraint,

$$E = \sum_i P_i C_i = \lambda \frac{\sigma-1}{\sigma} \sum_i \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = \lambda \frac{\sigma-1}{\sigma}. \quad (2.15)$$

Therefore the expenditure share is

$$s_i = \frac{P_i C_i}{E} = \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}}. \quad (2.16)$$

Rearranging,

$$\frac{P_i C_i}{E} = C_i^{\frac{\sigma-1}{\sigma}} C^{-\varepsilon_i \frac{\sigma-1}{\sigma}}, \quad (2.17)$$

$$\frac{P_i}{E} C_i^{1/\sigma} = C^{-\varepsilon_i \frac{\sigma-1}{\sigma}}, \quad (2.18)$$

$$C_i = \left(\frac{P_i}{E} \right)^{-\sigma} C^{\varepsilon_i(1-\sigma)}. \quad (2.19)$$

Equivalently,

$$s_i = \left(\frac{P_i C^{\varepsilon_i}}{E} \right)^{1-\sigma}. \quad (2.20)$$

2.3 Utility maximization

Given prices $P_i > 0$ and total expenditure $E > 0$, the consumer chooses consumption quantities, which determine the index C , subject to

$$\max_{\{C_i\}} C, \quad \sum_i P_i C_i = E, \quad \sum_i \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (2.21)$$

The Lagrangian is written as

$$\mathcal{L} = C + \rho \left[\sum_i \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} - 1 \right] + \lambda \left(E - \sum_i P_i C_i \right). \quad (2.22)$$

The commodity first-order conditions used to derive demand are

$$\frac{\partial \mathcal{L}}{\partial C_i} = \rho \frac{\sigma-1}{\sigma} \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{-1/\sigma} \frac{1}{C^{\varepsilon_i}} - \lambda P_i = 0, \quad (2.23)$$

$$\rho \frac{\sigma-1}{\sigma} \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = \lambda P_i C_i. \quad (2.24)$$

Summing over all i yields

$$\rho \frac{\sigma-1}{\sigma} \sum_i \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = \lambda \sum_i P_i C_i = \lambda E, \quad (2.25)$$

and hence

$$\rho \frac{\sigma-1}{\sigma} = \lambda E, \quad E = \frac{\rho}{\lambda} \frac{\sigma-1}{\sigma}. \quad (2.26)$$

The expenditure share therefore satisfies

$$s_i = \frac{P_i C_i}{E} = \left(\frac{C_i}{C^{\varepsilon_i}} \right)^{\frac{\sigma-1}{\sigma}} = \left(\frac{P_i C^{\varepsilon_i}}{E} \right)^{1-\sigma}. \quad (2.27)$$

As in the cost-minimization derivation,

$$\frac{P_i}{E} C_i^{1/\sigma} = C^{-\varepsilon_i \frac{\sigma-1}{\sigma}} \implies C_i = \left(\frac{P_i}{E} \right)^{-\sigma} C^{\varepsilon_i(1-\sigma)}. \quad (2.28)$$

2.4 NH-CES expenditure function

Using the optimal demands,

$$E = \sum_i P_i C_i \quad (2.29)$$

$$= \sum_i P_i \left(\frac{P_i}{E} \right)^{-\sigma} C^{\varepsilon_i(1-\sigma)} \quad (2.30)$$

$$= E^\sigma \sum_i P_i^{1-\sigma} C^{\varepsilon_i(1-\sigma)} = E^\sigma \sum_i (P_i C^{\varepsilon_i})^{1-\sigma}. \quad (2.31)$$

Therefore,

$$E(\mathbf{P}, C) = \left[\sum_i (P_i C^{\varepsilon_i})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (2.32)$$

2.5 NH-CES expenditure elasticity

2.5.1 Definition and derivation

Holding prices fixed, define the expenditure elasticity of good i by

$$\eta_i \equiv \frac{\partial \ln C_i}{\partial \ln E} = \frac{\partial \ln(P_i C_i)}{\partial \ln E} = \sigma + \varepsilon_i(1 - \sigma) \frac{\partial \ln C}{\partial \ln E}. \quad (2.33)$$

This follows by taking logarithms of optimal demand:

$$\ln C_i = \sigma(\ln E - \ln P_i) + \varepsilon_i(1 - \sigma) \ln C. \quad (2.34)$$

To calculate the remaining derivative, take logarithms of the expenditure function:

$$\ln E = \frac{1}{1 - \sigma} \ln \left[\sum_j P_j^{1-\sigma} C^{\varepsilon_j(1-\sigma)} \right]. \quad (2.35)$$

By the chain rule,

$$\frac{\partial \ln E}{\partial \ln C} = \frac{\partial \ln E}{\partial C} C \quad (2.36)$$

$$= \frac{C}{1 - \sigma} \frac{\sum_j P_j^{1-\sigma} \varepsilon_j(1 - \sigma) C^{\varepsilon_j(1-\sigma)-1}}{\sum_j P_j^{1-\sigma} C^{\varepsilon_j(1-\sigma)}} \quad (2.37)$$

$$= \frac{\sum_j \varepsilon_j (P_j C^{\varepsilon_j})^{1-\sigma}}{\sum_j (P_j C^{\varepsilon_j})^{1-\sigma}}. \quad (2.38)$$

Since

$$s_j = \frac{P_j C_j}{E} = \frac{(P_j C^{\varepsilon_j})^{1-\sigma}}{\sum_k (P_k C^{\varepsilon_k})^{1-\sigma}}, \quad (2.39)$$

define the expenditure-share-weighted average

$$\bar{\varepsilon} \equiv \sum_j s_j \varepsilon_j > 0. \quad (2.40)$$

It follows that

$$\frac{\partial \ln E}{\partial \ln C} = \bar{\varepsilon}, \quad \frac{\partial \ln C}{\partial \ln E} = \frac{1}{\bar{\varepsilon}}. \quad (2.41)$$

Substituting into (2.33) gives

$$\boxed{\eta_i = \sigma + (1 - \sigma) \frac{\varepsilon_i}{\bar{\varepsilon}}}. \quad (2.42)$$

2.5.2 Luxuries and necessities

At fixed prices, $\ln s_i = \ln(P_i C_i) - \ln E$, so

$$\boxed{\frac{\partial \ln s_i}{\partial \ln E} = \eta_i - 1 = (1 - \sigma) \frac{\varepsilon_i - \bar{\varepsilon}}{\bar{\varepsilon}}}. \quad (2.43)$$

A luxury has $\eta_i > 1$ and an increasing expenditure share. A good with $\eta_i < 1$ has a decreasing expenditure share. Within the latter category, a normal necessity satisfies $0 < \eta_i < 1$, whereas an inferior good satisfies $\eta_i < 0$. The case $\eta_i = 1$ has unit expenditure elasticity.

The parameter conditions for increasing or decreasing expenditure shares are as follows:

Parameter range	Luxury ($\eta_i > 1$)	Decreasing share ($\eta_i < 1$)
$0 < \sigma < 1$	$\varepsilon_i > \bar{\varepsilon}$	$\varepsilon_i < \bar{\varepsilon}$
$\sigma > 1$	$\varepsilon_i < \bar{\varepsilon}$	$\varepsilon_i > \bar{\varepsilon}$

When $0 < \sigma < 1$ and all $\varepsilon_i > 0$, all expenditure elasticities are positive; the decreasing-share category then consists of normal necessities. For $\sigma > 1$, positivity of η_i must be checked separately.

Interpretation. The classification depends on the consumer's expenditure composition. As expenditure changes, the shares s_j and hence $\bar{\varepsilon}$ change. In a system with at least three distinct nonhomotheticity parameters, a good with an intermediate parameter can move from luxury to necessity even though its own parameter remains constant. With two goods and distinct parameters, changing expenditure alone does not reverse their classification.

2.5.3 Engel aggregation

The share-weighted average of expenditure elasticities equals one. To see this, hold prices fixed and write

$$E = \sum_i P_i C_i(E). \quad (2.44)$$

Differentiating with respect to E ,

$$1 = \sum_i P_i \frac{\partial C_i}{\partial E} = \sum_i P_i \eta_i \frac{C_i}{E} = \sum_i s_i \eta_i. \quad (2.45)$$

The equality uses the elasticity identity

$$\eta_i = \frac{\partial \ln C_i}{\partial \ln E} = \frac{E}{C_i} \frac{\partial C_i}{\partial E} \implies \frac{\partial C_i}{\partial E} = \eta_i \frac{C_i}{E}. \quad (2.46)$$

Thus,

$$\boxed{\sum_i s_i \eta_i = 1.} \quad (2.47)$$

- A small 1% rise in total expenditure can generate different consumption growth rates across goods, but their initial-share-weighted average is approximately 1%.
- Not all goods with positive expenditure shares can simultaneously have elasticities strictly above one or strictly below one. All elasticities may equal one.

2.6 Euler equation under nonhomothetic CES

2.6.1 Household problem

The household chooses consumption C_t and next-period assets A_{t+1} :

$$\max_{\{C_t, A_{t+1}\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\theta}}{1-\theta}, \quad E_t(C_t) + A_{t+1} = W_t + R_t A_t. \quad (2.48)$$

Here A_0 is given, W_t is labor income, and

$$E_t(C_t) = \left[\sum_i (P_{it} C_t^{\varepsilon_i})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}. \quad (2.49)$$

Use $0 < \beta < 1$ and $\theta > 0$; the displayed utility formula assumes $\theta \neq 1$, with $u(C_t) = \ln C_t$ when $\theta = 1$. Define

$$s_{it} = \frac{P_{it}C_{it}}{E_t}, \quad \bar{\varepsilon}_t = \sum_i s_{it}\varepsilon_i, \quad P_t = \frac{E_t(C_t)}{C_t}. \quad (2.50)$$

2.6.2 Lagrangian and first-order conditions

The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[\beta^t \frac{C_t^{1-\theta}}{1-\theta} + \lambda_t (W_t + R_t A_t - A_{t+1} - E_t(C_t)) \right]. \quad (2.51)$$

The displayed first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial C_t} = \beta^t C_t^{-\theta} - \lambda_t E'_t(C_t) = 0, \quad (2.52)$$

$$\frac{\partial \mathcal{L}}{\partial A_{t+1}} = -\lambda_t + R_{t+1} \lambda_{t+1} = 0. \quad (2.53)$$

The second condition implies

$$\frac{\lambda_{t+1}}{\lambda_t} = \frac{1}{R_{t+1}}. \quad (2.54)$$

To evaluate $E'_t(C_t)$, start from

$$E_t^{1-\sigma} = \sum_i P_{it}^{1-\sigma} C_t^{\varepsilon_i(1-\sigma)}. \quad (2.55)$$

Differentiate both sides with respect to C_t :

$$(1-\sigma)E_t^{-\sigma} E'_t(C_t) = \sum_i \varepsilon_i (1-\sigma) P_{it}^{1-\sigma} C_t^{\varepsilon_i(1-\sigma)-1}. \quad (2.56)$$

Since

$$s_{it} = \frac{P_{it}^{1-\sigma} C_t^{\varepsilon_i(1-\sigma)}}{E_t^{1-\sigma}}, \quad (2.57)$$

we obtain

$$E'_t(C_t) = \frac{E_t^\sigma}{C_t} \sum_i \varepsilon_i P_{it}^{1-\sigma} C_t^{\varepsilon_i(1-\sigma)} \quad (2.58)$$

$$= \frac{E_t^\sigma}{C_t} \sum_i \varepsilon_i s_{it} E_t^{1-\sigma} \quad (2.59)$$

$$= \frac{E_t}{C_t} \sum_i \varepsilon_i s_{it} = P_t \bar{\varepsilon}_t. \quad (2.60)$$

Consequently, the consumption condition becomes

$$\boxed{\beta^t C_t^{-\theta} = \lambda_t \bar{\varepsilon}_t P_t.} \quad (2.61)$$

2.6.3 Euler equation

Taking the ratio of the consumption conditions at $t + 1$ and t gives

$$\beta \left(\frac{C_{t+1}}{C_t} \right)^{-\theta} = \frac{\lambda_{t+1}}{\lambda_t} \frac{\bar{\varepsilon}_{t+1}}{\bar{\varepsilon}_t} \frac{P_{t+1}}{P_t} = \frac{1}{R_{t+1}} \frac{\bar{\varepsilon}_{t+1}}{\bar{\varepsilon}_t} \frac{P_{t+1}}{P_t}. \quad (2.62)$$

Equivalently,

$$\boxed{\frac{C_t^{-\theta}}{\bar{\varepsilon}_t P_t} = \beta R_{t+1} \frac{C_{t+1}^{-\theta}}{\bar{\varepsilon}_{t+1} P_{t+1}}, \quad u'(C_t) = C_t^{-\theta}.} \quad (2.63)$$

Interpretation. At the optimum described by these conditions, the marginal utility of spending an extra dollar today equals the discounted marginal utility obtained by saving that dollar for tomorrow.

References

- [1] Comin, D., Lashkari, D., and Mestieri, M. (2021). Structural Change With Long-Run Income and Price Effects. *Econometrica*, 89(1), 311–374. doi:10.3982/ECTA16317.
- [2] Mestieri, M. (n.d.). *Step-by-step derivations for Nonhomothetic CES*. Teaching materials, available from the [author's research page](#).